

Andreas Tzitzikas

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Education

University of Maryland, College Park, MD	Expected May 2027
<i>Master of Science in Computer Engineering</i>	GPA: 4.00
<i>Bachelor of Science in Computer Engineering (GPA: 3.88)</i>	Expected May 2026
Relevant Coursework: Electric Circuits (ENEE205), Elements of Discrete Signal Analysis (ENEE222), Digital Circuits Lab (ENEE245), Introduction to Micro/Nanoelectronics (ENEE304), Microprocessors (ENEE440)	

Projects

STM32G4 Bare-Metal Peripheral Integration ARM Assembly, Analog/Digital Interfaces	Jan 2025 – May 2025
- Implemented direct register-level programming for ADC, DAC, PWM/PFM generation using TIM2, and SPI flash memory interfaces on STM32G4 microcontroller platform. - Engineered multiplexed interrupt system routing single hardware interrupt to multiple peripheral drivers with state-based arbitration for efficient resource utilization. - Developed comprehensive analog signal processing including ADC sampling, DAC output generation, and PWM motor control with precise timing and synchronization requirements.	
PCB Design with RP2350 Microcontroller KiCad, Power Systems, Signal Integrity	Jan 2024 – Present
- Designed complete PCB system with QFN60 package RP2350 microcontroller, implementing proper power distribution networks, decoupling capacitors, and signal integrity for reliable high-speed operation. - Developed sensor interface circuits with appropriate filtering, amplification, and protection circuitry for robust analog signal acquisition and processing. - Generated fabrication documentation including layer stackup specifications, Gerber files, bill of materials, and assembly guidelines for professional PCB manufacturing.	
Tomasulo Out-of-Order RISC-V CPU Digital Design, Timing Analysis, Verification	Nov 2025 - Dec 2025
- Architected out-of-order execution system using Tomasulo's Algorithm with register renaming, reservation stations, and reorder buffer for advanced digital system design. - Achieved 36% higher operating frequency (69.7MHz vs 51.2MHz) and 40-150% IPC improvement compared to in-order implementation through timing optimization and performance analysis. - Synthesized to fabrication-ready GDSII using Sky130 130nm PDK, achieving zero DRC/LVS errors and comprehensive static timing analysis for reliable digital circuit operation.	

Experience

Embedded Systems Intern, Alchemy – College Park, MD	Jun 2025 – Aug 2025
- Engineered real-time control systems for hybrid solid oxide fuel cell reactors, implementing precise analog signal processing and motor control with PWM and ADC interfaces. - Developed embedded firmware with interrupt-driven architectures, real-time scheduling, and robust error handling for industrial control applications. - Integrated analog sensors and actuators with digital control systems, ensuring signal integrity, noise immunity, and reliable operation in harsh environments.	
Undergraduate Teaching Assistant, ENEE205 Electric Circuits – College Park, MD	Sep 2025 – Present
- Lead laboratory and discussion sections covering circuit theory fundamentals, Kirchhoff's laws, Thevenin/Norton equivalents, and AC/DC circuit analysis. - Provide hands-on guidance with oscilloscopes, signal generators, and circuit simulation tools for practical circuit design and measurement techniques.	
Technical Coordinator, Electronics Prototyping Lab, Terrapin Works – College Park, MD	Jan 2024 – Present
- Maintain and operate LPKF PCB prototyping equipment ensuring reliable fabrication capabilities with 90% uptime for student and faculty projects. - Provide technical support for PCB design, fabrication, and assembly processes including schematic capture, layout, and component soldering techniques.	

Skills

Circuit Design: Analog/Digital Circuits, Signal Processing, Power Systems, Filter Design
Electronics: Microcontrollers, ADC/DAC, PWM, SPI/I2C/UART, GPIO, Embedded Programming
Laboratory Tools: Oscilloscopes, Signal Generators, Multimeters, PCB Fabrication
Software Tools: KiCad, Cadence Spectre, MATLAB, LTspice, Xilinx Vivado
Analysis: Circuit Analysis, Signal Processing, Control Systems, Electromagnetic Compatibility